

ECE 315 – Signals and Systems, Spring 2026

Midterm Exam II

April 17, 2026

Exam duration: 50 minutes

INSTRUCTIONS

1. This is a **closed-book** examination.
2. The exam has **four** questions and comprises **seven** pages.
3. Show all work for full credit.
4. Students may use scientific calculators, but wireless communication is prohibited. Other electronic devices (e.g., computers, phones, tablets, digital notebooks, etc.) are not allowed.
5. The mathematical formulas below may be helpful.

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$$

$$\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

$$\sin(x + y) = \sin(x) \cos(y) + \cos(x) \sin(y)$$

$$\cos(x + y) = \cos(x) \cos(y) - \sin(x) \sin(y)$$

$$\sin(x) \cos(y) = \frac{1}{2}(\sin(x - y) + \sin(x + y))$$

$$\sum_{n=0}^{N-1} \alpha^n = \frac{1-\alpha^N}{1-\alpha} \quad \text{for } \alpha \neq 1$$

$$\sum_{n=0}^{\infty} \alpha^n = \frac{1}{1-\alpha} \quad \text{for } |\alpha| < 1$$

Question 1

(4pts) Consider the following discrete-time periodic signal:

- $x[n]$ is periodic with period 8 and
$$x[n] = \begin{cases} 1 & \text{when } |n| \leq 1 \\ 0 & \text{when } 2 \leq |n| \leq 4 \end{cases}$$

Compute its Fourier series representation, and select which types of convergence and justify your choice:

- ☐ Convergence everywhere
- ☐ Convergence with no energy difference
- ☐ Convergence except for discontinuities

Question 2

(4pts) Prove the Parseval's theorem for continuous-time periodic signals. I.e., show the following equality for a continuous-time periodic signal $x(t)$ with period T .

$$\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

where a_k is the Fourier series coefficients of $x(t)$.

Question 3

(6pts) Assume that we observe the following properties for a continuous-time signal $x(t)$:

- $x(t)$ is a real signal;
- $x(t)$ is periodic with period $T = 2$;
- $x(t)$ has Fourier series coefficients a_k that satisfy: (i) $a_k = 0$ for $|k| > 1$;
(ii) The signal with Fourier series coefficients $b_k = e^{-j\pi k} a_{-k}$ is odd;
- $\frac{1}{2} \int_2 |x(t)|^2 dt = 1$.

Determine the signal $x(t)$.

Question 4

(6pts) Consider a discrete-time linear-time invariant (LTI) system with impulse response $h[n] = \frac{1}{2^n}u[n-1]$, and an input $x[n]$,

$$x[n] = 1 + \cos\left(\frac{\pi}{4}n\right).$$

- (i) Determine the system function (system's eigenvalue) $H(z)$ and its frequency response $H(e^{jw})$;
- (ii) Compute the output of the LTI system for the input $x[n]$.

